

# CRESS: Cross-Category Reliability Estimation with Source Support for Low-Storage Few-Shot Industrial Anomaly Detection

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## ABSTRACT

Few-shot industrial anomaly detectors built on frozen foundation features often provide strong anomaly rankings, but their thresholds may not maintain consistent false-alarm rates across categories or input shifts. To address this, we audit reliability separately from ranking. Our framework combines a low-storage DINOv2 PCA residual with label-free leave-one-image-out (LOIO) p-values from  $k$  normal supports. Evaluating 15 MVTec and 12 VisA categories under four corruption types, our audit reveals a resolution floor: target-only alarms cannot trigger below  $\alpha = 1/(k+1)$ . Empirical normal p-values range from conservative on VisA at  $k = 4$  to anti-conservative on corrupted MVTec. We further introduce Cross-category Reliability Estimation with Source Support (CRESS), a source-assisted procedure that separates reference, threshold-proposal, and certification categories. Across a frozen grid of four  $k$  values, five seeds, five conditions, within-dataset tests, and MVTec-to-VisA/MPDD transfer, strict category-level certification returns the fail-closed zero threshold in all 960 gate cells. This outcome is not solely attributable to Hoeffding looseness: when all observed category losses are zero, even an optimistic multiplicity-free, deterministic, uniformly valid, distribution-free 95% upper bound requires at least 14 independent categories to certify  $\alpha = 0.20$ . The frozen protocol requires more, whereas only three or four certification categories are available. Fixed-archive image certificates select positive thresholds in 37-60% of settings, but estimate image-level risk under the observed source mixture rather than new-category risk. Historical non-independent CRESS results are therefore retained only as empirical diagnostics. Our contribution is an auditable framework that separates empirical threshold utility from transferable certification and quantifies when few-category source archives cannot support non-vacuous reliability claims, without claiming target-category control unsupported by the available evidence.

## 1. Introduction

Industrial anomaly detection (AD) is commonly implemented in asymmetric data regimes, where normal examples are available, while abnormal examples are rare, heterogeneous, and expensive to annotate. Consequently, few-shot AD has become practically significant. In this context, an inspection system typically receives only  $k$  normal support images for a target category, often with  $k \in \{1, 2, 4, 8\}$ , and is required to detect both image-level and pixel-level defects during testing.

Recent approaches have improved performance in this regime by leveraging frozen visual foundation models. For example, DINOv2 [?] patch features can be stored in a normal memory bank and queried using nearest-neighbor distances, as implemented in PatchCore-style methods and AnomalyDINO [? ?]. Alternatively, frozen patch features may be compressed into a normal principal component analysis (PCA) or subspace model, as in SubspaceAD [1]. These methods enable effective anomaly ranking without the need to train a large task-specific backbone.

However, effective ranking does not equate to reliable decision support. A raw nearest-neighbor distance or PCA residual may yield high area under the receiver operating

characteristic curve (AUROC) while remaining poorly calibrated: a score threshold that is effective for one class may not correspond to the same false-alarm rate for another class, dataset, or under data corruption. This distinction is critical in industrial deployment, where operators require not only identification of suspicious samples but also trustworthy alarm probabilities. Previous studies have demonstrated that DINOv2-based few-shot AD can exhibit poor calibration and adversarial fragility [2]. As such, reliability is an integral aspect of the deployment challenge, rather than a superficial post-processing concern.

This work addresses the identified gap by decoupling anomaly ranking from reliability estimation. The central questions pertain to the learning system itself: what false-alarm behavior can a decision rule, calibrated on  $k$  examples of a frozen representation, guarantee, at what resolution, and how does this guarantee degrade under input distribution shifts? The ranking component utilizes a frozen DINOv2 PCA/subspace detector, aggregating patch residuals into a raw anomaly score for AUROC, average precision (AP), and localization. A distinct reliability component transforms this evidence into calibrated or probability-like outputs. The primary target-only approach employs leave-one-image-out (LOIO) conformal calibration on the limited normal support images. Each support image is evaluated as if held out, generating normal nonconformity values; a test image subsequently receives a conformal p-value and a reliability score of  $1 - p$ . This framework does not constitute a supervised

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anomaly posterior, but it is statistically interpretable and constructed without target anomaly labels.

The foundational components utilized in this study—DINOv2 features, PCA residuals, Platt-family calibration, and conformal anomaly detection—are well-established in the literature; the present contribution builds upon these elements. However, existing research leaves unresolved structural questions specific to the few-shot regime: for example, a conformal alarm calibrated on  $k$  support images cannot trigger below  $\alpha = 1/(k+1)$ , corruption shift can render the observed false-alarm rate anti-conservative, and a statistically valid source-category certificate may be vacuous when only a limited number of categories are available. The primary contribution integrates an exact resolution-floor diagnosis, an empirical audit under distribution shift, and **Cross-category Reliability Estimation with Source Support (CRESS)**, a source-assisted thresholding and certification procedure evaluated using a strict nested protocol. The rigorous experimental protocol does not support the anticipated new-category claim: all category-certified thresholds revert to zero. This negative outcome is consistent with both the stated Hoeffding rule and a broader all-zero lower bound for deterministic, distribution-free category-risk confidence bounds, and is reported alongside, rather than in place of, fixed-archive image sensitivity and historical empirical results.

The implications for learning systems are broadly applicable, despite the industrial testbed: finite calibration constrains both the resolution of target-only alarms and the feasibility of source-category certification. This paper presents four contributions, with the first two being central:

- We make the few-shot resolution limit of conformal alarms explicit through an attainable-alpha analysis: with  $k$  supports, no alarm can trigger below  $\alpha = 1/(k+1)$ , and empirical false-alarm rates must be interpreted relative to the attainable grid rather than the nominal level. Cluster-aware Monte Carlo diagnostics of normal p-values then localize when and in which direction the exchangeable-null pattern fails under corruption shift, being conservative on VisA and anti-conservative on corrupted MVTec.
- We formulate CRESS with disjoint reference, proposal, and certification categories and evaluate it under a strict nested protocol. In addition to the Hoeffding-specific feasibility condition, we prove that any deterministic uniformly valid distribution-free upper bound based solely on independent and identically distributed (iid) category losses has an all-zero floor  $1 - \beta^{1/n}$ ; even without multiplicity, 95% certification at  $\alpha = 0.20$  requires 14 categories. The frozen experiment spans four  $k$ , five random seeds, five conditions, MVTec, VisA, and MVTec-to-VisA/MPDD transfer. Category certification yields a zero threshold in all 960 gate cells because only three

or four certification categories are available; image-level fixed-archive sensitivity is reported separately and never relabeled as a new-category result.

- We formulate a decoupled pipeline: frozen DINOv2 PCA residuals are used for ranking, while a label-free layer provides Vector/Shift-Aware calibration and LOIO p-values [3] based on the normal supports. Across the historical MVTec and VisA corruption studies, LOIO reduces expected calibration error (ECE) compared to most tested calibrators without altering the raw ranking; operational metrics, rather than ECE, carry the central interpretation.
- We report accuracy-storage, transfer calibration, pixel-level, robustness, prevalence-stress, and routing diagnostics, including negative findings that prevent overclaiming. The framework is not MVTec AU-ROC SOTA, nor is it adversarially robust; ECE on conformal-derived scores is strongly prevalence-sensitive, and the historical CRESS gate is not an independent certificate.

The remainder of this paper is structured as follows. Section 2 positions the work against few-shot detectors, calibration, and conformal anomaly detection. Section 3 details the decoupled pipeline, the LOIO conformal route, the attainable-alpha analysis, and CRESS. Section 4 outlines datasets, protocols, and claim rules. Section 5 presents calibration comparisons, uniformity tests, operational false-alarm behavior, CRESS evaluations with their pre-specified gate, and diagnostic analyses. Section 6 discusses limitations, and Section 7 concludes.

## 2. Related Work

### 2.1. Few-Shot Industrial Anomaly Detection

Classical industrial anomaly detection (AD) typically assumes access to normal-only training data and identifies deviations at test time. Patch distribution, self-supervised, and reconstruction-discriminative approaches such as PaDiM, CutPaste, and DRAEM provide important context for this literature [4–6]. PatchCore [?] established memory-bank patch retrieval as a robust baseline by storing representative normal features and scoring test patches using nearest-neighbor distance. AnomalyDINO [?] extends this paradigm to few-shot inspection with frozen DINOv2 representations. The few-shot AD literature has further introduced registration-based adaptation (RegAD [7]), graph-structured memory (GraphCore [8]), fast feature reconstruction (FastRecon [9]), vision-language zero- and few-shot scoring (WinCLIP [10]; CLIP-SAM collaboration [11] and cross-modal prompt regularization [12] continue this direction), and in-context residual learning (InCTRL [13]). While these methods are effective rankers, the reference state can remain nontrivial at inference time, and the raw score scale does not represent a calibrated probability. Calibration under distribution shift is also known to degrade for deep models generally [14]. Our work is complementary: it does not

compete on ranking but instead examines how few-shot detectors should expose decision reliability.

## 2.2. Frozen-Feature Subspace Detectors

Subspace approaches replace explicit patch retrieval with a compact normal subspace. SubspaceAD [1] demonstrates that frozen DINOv2 patch features and PCA or subspace residuals are already effective for few-shot anomaly ranking, establishing the ranking mechanism as settled prior art. Our contribution begins after the residual is computed, focusing on how a low-storage subspace detector should expose reliability under low-shot and distribution shift conditions, as well as the guarantees its alarms can provide.

## 2.3. Calibration and Robustness in Few-Shot AD

Calibration evaluates whether predicted confidence aligns with empirical correctness [15, 16]. Standard post-hoc calibrators include Platt scaling [15], temperature scaling [16], histogram binning [17], and isotonic regression [18]; all are compared under a shared label-free protocol. In anomaly detection, calibration is challenging due to sparse labels, open-ended anomaly types, and varying score distributions across classes. Reliability analyses of DINOv2-based few-shot AD report poor expected calibration error (ECE) and significant FGSM fragility [2]. We therefore evaluate ECE, Brier score, and negative log-likelihood (NLL) are evaluated separately from AUROC and average precision (AP), and adversarial results, including FGSM-style perturbations [19], are reported as diagnostics rather than as robustness improvements.

## 2.4. Conformal Prediction and Reliability Routing

Conformal prediction transforms nonconformity scores into p-values under exchangeability assumptions [20–22]. This approach is particularly suitable for normal-only few-shot AD, as support normal images can define a reference distribution without requiring real anomalies. Conformal anomaly detection is well established: leave-one-out, bootstrap, and cross-conformal anomaly detectors have been directly analyzed [3]; weighted conformal detection in low-data regimes, including resolution and effective-sample-size analyses, has been developed [23]; general nonconformity toolkits are available [24]; and few-shot conformal prediction with auxiliary tasks addresses the small-calibration-set problem [25]. Conformal AD, LOIO calibration, and weighted conformal p-values are thus established mechanisms. The present contribution begins where these methods reach their limits in the few-shot regime: at the resolution floor  $1/(k+1)$  that no target-only  $k$ -shot conformal alarm can surpass, and at the independent-category requirement of a simultaneous source certificate. This work proves the resolution barrier and both Hoeffding-specific and distribution-free all-zero category-feasibility bounds, empirically audits corruption-shift deviations, and evaluates CRESS thresholding below the target-only floor without target anomaly labels. Gated-expert models, including SAGE-style shared and specialized routing [26], motivate the broader concept of selecting conservative or shift-specialized reliability paths.

The proposed framework adopts this routing perspective, but its primary target-only route is intentionally simple: fixed LOIO conformal reliability. Learned or weighted gates are considered ablations unless they improve reliability without test-label expert selection.

## 3. Method

### 3.1. Problem Setup

For each category  $c$ , we observe a support set  $\mathcal{S}_c = x_{i1} = 1^k$  containing only normal images. The test set includes both normal and anomalous images, but anomaly labels are reserved exclusively for evaluation and are not utilized in fitting the main detector or calibrator. The objectives are to generate: (i) a raw anomaly ranking score  $s(x)$ , (ii) a probability-like reliability score  $\hat{p}(y = 1 | x)$  or conformal anomaly confidence, and (iii) optional pixel-level maps. Figure 1 provides an overview: a frozen backbone supplies a ranking lane that remains unaltered, a label-free target-only reliability lane whose conformal alarms reach the attainable-alpha floor, and a CRESS lane that attempts source-assisted operation below that floor but yields a zero threshold when certification is inadequate.

### 3.2. Frozen Patch Features and Subspace Ranking

We extract patch features with a frozen DINOv2 ViT-S/14 backbone. Let  $z_{ij} \in \mathbb{R}^d$  denote patch  $j$  from image  $i$ . A PCA/subspace model is fit on support patches. With support mean  $\mu$  and retained principal directions  $U \in \mathbb{R}^{d \times m}$ , the patch residual is

$$r(z) = \left| (z - \mu) - UU^T(z - \mu) \right|_2. \quad (1)$$

Patch residuals form an anomaly map. The image-level ranking score is a high-residual aggregation,

$$s(x) = \text{Agg } j, r(z_j), \quad (2)$$

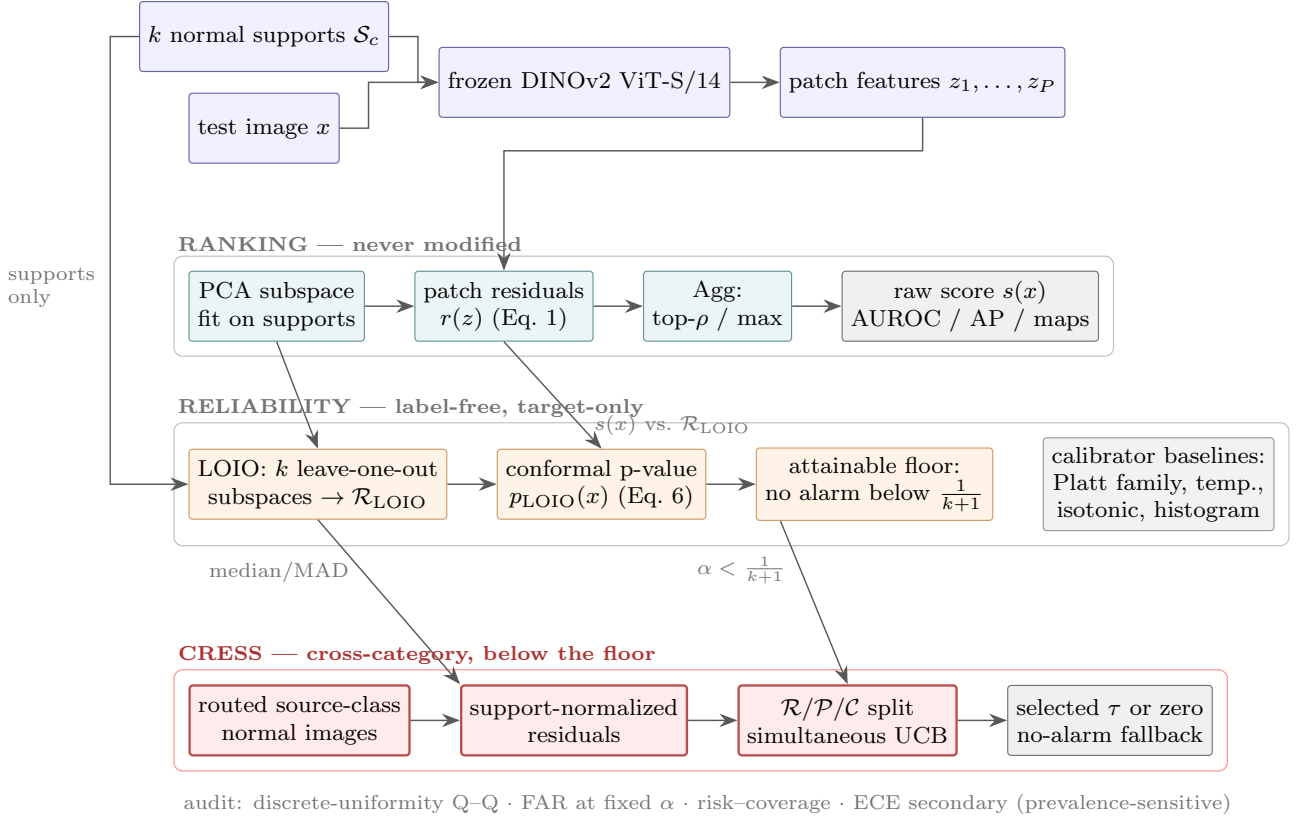
where Agg denotes the patch maximum in the clean accuracy-storage benchmark and the mean of the top-1% patch residuals ( $\rho=0.01$ ) in the conformal pipeline; the top-fraction variant serves as a smoothed maximum to stabilize LOIO calibration scores. We state this explicitly because the two protocols employ different aggregation methods, and Section 5 reports an aggregation-sensitivity ablation. Notably, this raw score is the quantity used for AUROC and AP, and calibration procedures do not alter the ranking score.

### 3.3. Vector and Shift-Aware Calibration

The first reliability family is post-hoc logistic calibration. A scalar Platt scaler maps  $s(x)$  to a probability, while Vector Platt uses a richer descriptor

$$\phi(x) = [s_{\text{pca}}(x), s_{\text{head}}(x), |\tilde{s}_{\text{pca}}(x) - \tilde{s}_{\text{head}}(x)|], \quad (3)$$

where  $s_{\text{head}}$  is a lightweight synthetic-anomaly head score and tildes denote normalized scores. The head is a small MLP ( $384 \rightarrow 256 \rightarrow 1$ , dropout 0.1) trained with binary cross-entropy to separate support normal patches from synthetically perturbed patches; a patch perturbation resamples a



**Figure 1:** Reliability-framework overview. A frozen DINOv2 backbone feeds two decoupled paths: the *ranking* lane (PCA residuals, never modified; used for AUROC/AP and localization) and the label-free target-only *reliability* lane (LOIO conformal p-values from the  $k$  supports, with Platt-family and standard scalar calibrators as baselines). Target-only alarms are structurally silent below  $1/(k+1)$ . The strict CRESS lane pools routed source-class normals, separates reference, proposal, and certification classes, and selects a threshold only when its simultaneous source-domain upper bound passes; otherwise it returns a no-alarm threshold.

support patch and adds Gaussian noise plus a sign shift in feature space. Its image-level score is the maximum patch probability. The head participates only in the calibration descriptor  $\phi(x)$ , never in the anomaly ranking. The calibrated probability is

$$\hat{p}_\theta(x) = \sigma(w^\top \phi(x) + b). \quad (4)$$

Shift-Aware Platt augments  $\phi(x)$  by incorporating five shift descriptors: the maximum and mean distance of test patches to the support feature center (scaled by the support 95th-percentile norm), the mean and standard deviation of patch residual scores, and a residual concentration ratio (maximum over mean). Weighted Platt reweights calibration samples by density-ratio estimates from a logistic domain classifier, following weighted conformal prediction under covariate shift [27]. While these variants are effective for structured shifts, they may over-adapt, thereby motivating the adoption of a conservative conformal route.

*Calibrator training protocol and label source.* All calibrators in the primary protocol are trained without real anomaly labels (mode `normal_synthetic`). The calibration set consists of the  $k$  support normal images (label 0) and an equal number of synthetic images (label 1) obtained by

replacing a random 25% of a support image’s patch features with perturbed support patches as described above. Vector-family calibrators fit a multivariate logistic model on  $\phi(x)$  using gradient descent with  $L_2$  regularization. The memory-bank and plain-subspace baselines, which do not utilize a calibration descriptor, apply scalar Platt scaling to the raw score under the same synthetic protocol. Standard scalar calibrators evaluated in Section 5 - temperature scaling [16], isotonic regression [18], and histogram binning [17] - are fit per class on the identical calibration set, mapping the raw score  $s(x)$  to a probability. Since raw residuals are not logits, temperature scaling is applied after centering at the midpoint of the two calibration class means,  $\hat{p}(x) = \sigma((s(x) - c)/T)$ , with only  $T$  learned. An `anomaly_val` mode that calibrates on 10% held-out real anomalies is reported solely as an explicitly labeled upper bound. This symmetric, label-free protocol ensures a fair LOIO-versus-calibrator comparison: every family sees exactly the same  $k$  support normals and no test information.

### 3.4. LOIO Conformal Reliability

For  $k \geq 2$ , the target-only route constructs normal calibration residuals by leave-one-image-out support splitting. For each support image  $x_i$ , a subspace is fit on  $\mathcal{S}_c \setminus \{x_i\}$



and the held-out image receives a residual score  $s_{-i}(x_i)$ . The resulting set

$$\mathcal{R}_{\text{LOIO}} = \{s_{-i}(x_i) : x_i \in \mathcal{S}_c\} \quad (5)$$

serves as normal nonconformity evidence. For a test image  $x$ , the conformal p-value is

$$p_{\text{LOIO}}(x) = \frac{1 + \sum_{r \in \mathcal{R}_{\text{LOIO}}} \mathbf{1}\{r \geq s(x)\}}{1 + |\mathcal{R}_{\text{LOIO}}|}. \quad (6)$$

We report  $p_{\text{conf}}(x) = 1 - p_{\text{LOIO}}(x)$  as a probability-like anomaly reliability view. It should be interpreted as relative extremeness against held-out support normals, not as a fully supervised posterior probability. Weighted conformal replaces the uniform count by density-ratio weights [27] and reports effective sample size; empirically, it is less stable than LOIO on the main benchmark.

*Exchangeability caveat.* The construction above is deliberately asymmetric: each calibration residual  $s_{-i}(x_i)$  comes from a subspace fit on  $k-1$  images, while the test score  $s(x)$  uses all  $k$ . The two statistics are therefore not exactly exchangeable, and Eq. (6) is a leave-one-out approximation in the spirit of cross-conformal and jackknife+ inference [3, 20, 28] rather than a split-conformal p-value with a formal finite-sample guarantee [29, 30]. We handle this empirically on two fronts. First, a matched-LOIO audit scores held-out support and test images under the same fold-specific subspaces (pairing  $s_{-i}(x)$  with  $r_{-i}$ , the jackknife+ pairing): it does not improve the operational false-alarm/power trade-off (Section 5), so the simpler construction is retained. Second, rather than assuming validity, we audit it directly: Section 5 reports discrete-grid diagnostics of the normal p-values. Their reference distribution is simulated under an idealized exchangeable null while preserving shared-calibration dependence within class-seed clusters; it localizes empirical deviations but is not proof of exchangeability.

### 3.5. Attainable Alpha and the Few-Shot Resolution Floor

**Proposition 1 (attainable-alpha floor).** For any finite calibration scores  $r_1, \dots, r_k$ , a p-value of the form in Eq. (6) belongs to  $\{j/(k+1) : j = 1, \dots, k+1\}$ . Hence the alarm  $\mathbf{1}\{p(x) \leq \alpha\}$  is identically zero for  $\alpha < 1/(k+1)$ .

*Proof.* The exceedance count  $\sum_i \mathbf{1}\{r_i \geq s(x)\}$  is an integer between 0 and  $k$ . Adding one and dividing by  $k+1$  gives the stated grid and lower bound. No exchangeability assumption is needed for this algebraic claim.  $\square$

For  $k=4$ , this floor is 0.2; for  $k=8$ , it is  $1/9 \approx 0.111$ . The resolution limit is made explicit through an attainable-alpha analysis, which, for each nominal  $\alpha$ , reports the nearest attainable grid point and the corresponding empirical false-alarm rate. This structural property delineates the regime where cross-category support may be beneficial; it is not claimed as an algorithmic innovation. The standard approach to address discreteness is a smoothed (randomized) conformal p-value, which employs an independent uniform tie-break and achieves uniformity under the relevant exchangeability assumptions. Section 5 evaluates this method as a

baseline, noting that randomization overcomes the floor but does not resolve the observed support–test shift.

### 3.6. Cross-Category Reliability Estimation with Source Support (CRESS)

category settings, normal images from *other* categories (source classes) are abundant, even when the target category contains only  $k$  normal supports. CRESS leverages these source images without requiring target anomaly labels. To ensure comparability, target and source residuals are normalized: each raw score is centered and scaled using the robust median and MAD statistics of its respective class’s LOIO support residuals. This normalization serves as an empirical comparability mechanism rather than evidence that class distributions coincide. In matched-condition mode, source images share the target’s corruption condition, necessitating condition metadata. Clean-source mode exclusively utilizes clean normals. The condition-agnostic ablation computes the median score across all available condition views for each base source image, ensuring that repeated corruptions are not treated as independent observations; deterministic mismatched-condition routing functions solely as a negative control. Pooling source normals enables finer p-value resolution than the  $k$  target supports and allows for thresholds below  $1/(k+1)$ .

### 3.7. Nested Source Certification and Transfer Boundary

For a fixed target, source classes are partitioned before certification into reference classes  $\mathcal{R}$ , proposal classes  $\mathcal{P}$ , and certification classes  $\mathcal{C}$ . Scores from  $\mathcal{R}$  define conformal p-values. The distinct p-values on  $\mathcal{P}$  propose at most  $M$  positive candidate thresholds. For candidate  $\tau$ , certification unit  $i$  has a loss  $L_i(\tau) \in [0, 1]$ . An image unit uses its Bernoulli alarm indicator and targets risk for the fixed source-image mixture. A class unit averages alarms within a held-out source class and targets a new draw from a source-category meta-population. Let  $A$  be the number of reported  $\alpha$  levels. For the class unit with  $n$  classes, CRESS uses

$$U_H(\tau) = \min \left\{ 1, \bar{L}(\tau) + \sqrt{\frac{\log(2AM/\delta)}{2n}} \right\}. \quad (7)$$

For independent Bernoulli image alarms, we use the exact one-sided binomial (Clopper-Pearson) upper bound with tail probability  $\delta/(2AM)$ . The factor  $2A$  allocates the family error across both units and all reported levels;  $M$  allocates it across candidates. CRESS selects the largest proposed  $\tau$  whose corresponding upper bound is at most  $\alpha$ . If none passes,  $\tau = 0$  gives a deterministic no-alarm fallback. Reference, proposal, and certification classes are disjoint; target test images and labels enter none of these stages.

**Proposition 2 (conditional source certificate).** Conditional on the reference and proposal stages, suppose certification units are independent draws from the declared source-unit population: Bernoulli losses for the image analysis, or bounded losses in  $[0, 1]$  for the class analysis. Then, with probability at least  $1 - \delta$ , simultaneously across the  $A$  levels

and both unit definitions, the corresponding source risk of every selected threshold is at most its  $\alpha$ .

*Proof.* For a fixed level, unit, and candidate, the one-sided Clopper-Pearson construction or Hoeffding's inequality fails with probability at most  $\delta/(2AM)$ . A union bound over  $M$  candidates,  $A$  levels, and two units gives total failure probability at most  $\delta$ ; no independence between these bounds is required. Selection occurs only on this simultaneous event, so each passing candidate has risk at most its level. The zero fallback has zero alarm risk.  $\square$

**Proposition 3 (Hoeffding feasibility).** Under the category-level Hoeffding rule above, a positive candidate can pass at level  $\alpha$  only if

$$n \geq \frac{\log(2AM/\delta)}{2\alpha^2}. \quad (8)$$

*Proof.* Because  $\bar{L}(\tau) \geq 0$ , the unclipped upper bound is at least  $\sqrt{\log(2AM/\delta)/(2n)}$ . Requiring it to be no larger than  $\alpha$  and rearranging gives Eq. (8). Any positive empirical loss only increases the requirement.  $\square$

**Proposition 4 (distribution-free all-zero lower bound).** Let  $\beta \in (0, 1)$  and let  $L_1, \dots, L_n$  be iid losses from an arbitrary distribution  $P$  on  $[0, 1]$ , with mean  $\mu_P$ . Let  $U_n : [0, 1]^n \rightarrow [0, 1]$  be any deterministic upper confidence bound satisfying

$$\inf_P \Pr_{P^n} \{ \mu_P \leq U_n(L_1, \dots, L_n) \} \geq 1 - \beta.$$

Then

$$U_n(0, \dots, 0) \geq 1 - \beta^{1/n}. \quad (9)$$

Consequently, even after observing zero loss in every certification category,  $U_n \leq \alpha$  is possible only if

$$n \geq \frac{\log \beta}{\log(1 - \alpha)}.$$

*Proof.* Write  $u = U_n(0, \dots, 0)$  and suppose, for contradiction, that  $u < 1 - \beta^{1/n}$ . Choose a Bernoulli distribution with mean  $q$  such that  $u < q < 1 - \beta^{1/n}$ ; Bernoulli laws are a subclass of distributions on  $[0, 1]$ . Its all-zero sample has probability  $(1 - q)^n > \beta$ . On that event  $U_n = u < q = \mu_P$ , so the bound fails with probability greater than  $\beta$ , contradicting uniform  $(1 - \beta)$  coverage. Rearranging  $1 - \beta^{1/n} \leq \alpha$  gives the sample-size condition.  $\square$

Proposition 4 shows that the frozen failure is not solely an artifact of Hoeffding's looseness. Even for one fixed candidate, one level, no image-unit co-reporting, and no multiplicity penalty ( $\beta = \delta = 0.05$ ), any deterministic uniformly valid distribution-free bound based only on iid category losses needs at least 14, 29, and 59 categories at  $\alpha = 0.20, 0.10$ , and  $0.05$ . Under the frozen equal allocation  $\beta = \delta/(2AM)$  with  $A = 3$  and the most favorable  $M = 1$ , the corresponding lower limits are 22, 46, and 94; the declared Hoeffding rule requires 60, 240, and 958. Every requirement exceeds the available three or four certification categories.

**Table 1**

Best-case minimum independent certification categories at  $\delta = 0.05$ , assuming zero observed category loss. "DF floor" is the distribution-free lower bound in Proposition 4: first without multiplicity ( $\beta = \delta$ ), then under the frozen allocation  $\beta = \delta/(2AM)$  with  $A = 3$ . "Hoeffding" is Proposition 3.

| rule / candidates         | $\alpha = .20$ | $\alpha = .10$ | $\alpha = .05$ |
|---------------------------|----------------|----------------|----------------|
| DF floor, no multiplicity | 14             | 29             | 59             |
| DF floor, $M = 1$         | 22             | 46             | 94             |
| DF floor, $M = 5$         | 29             | 61             | 125            |
| DF floor, $M = 20$        | 35             | 74             | 152            |
| Hoeffding, $M = 1$        | 60             | 240            | 958            |
| Hoeffding, $M = 5$        | 80             | 320            | 1,280          |
| Hoeffding, $M = 20$       | 98             | 390            | 1,557          |

Table 1 presents a design calculation rather than an empirical result. Proposition 4 is intentionally limited in scope: it addresses deterministic, uniformly valid, distribution-free upper bounds that rely solely on iid bounded category losses. Parametric or hierarchical assumptions, informative side data, or randomized confidence procedures are excluded from its statement and would necessitate separate validity arguments. This elementary minimax argument is not introduced as new confidence-bound theory; its purpose is to preclude the misinterpretation that the observed phenomenon is solely due to the looseness of Hoeffding's inequality. The calculation further demonstrates that adding source images or repeated support seeds cannot compensate for a lack of independent categories: such additions may refine the fixed-archive image analysis but do not increase the category-level  $n$ . Additionally, altering the certification unit after observing the result would modify the estimand and does not constitute a valid adjustment to the frozen protocol.

**Corollary 1 (conditional target transfer).** In the event described in Proposition 2, the target risk is also at most  $\alpha$  if, for every proposed threshold, the target normal alarm probability is no larger than the corresponding source-unit risk. The same conclusion follows if the target category is exchangeable with the certification-category meta-population.

The corollary's dominance or category-exchangeability condition is an assumption, not a consequence of robust normalization or matched routing. The two source certificates answer different questions: an image certificate for a fixed archive mixture cannot be relabeled a new-category guarantee, while a class certificate with few categories may be vacuous and cannot be replaced by the image result. The frozen nested results in Section 5 exhibit exactly this category-level failure; the older non-independent results remain labeled historical empirical diagnostics. A direct pooled-source conformal baseline uses the entire routed source pool with threshold  $\alpha$  and receives no artificial data restriction; unlike CRESS, it performs no proposal/certification step.

The broader routing component draws inspiration from the shared-versus-specialized principle in gated-expert systems. Candidate reliability views include Vector Platt, Shift-Aware Platt, weighted Platt, LOIO conformal, and weighted

conformal. We evaluate fixed routes, validation-ECE selection, and simple no-label rules based on shift descriptors and effective sample size. Because learned gates did not consistently outperform fixed LOIO across all settings, the main paper designates fixed LOIO as the target-only reference and treats learned gates as diagnostic ablations; these are not characterized as safe routes under distribution shift.

### 3.8. Selective Diagnostics

In addition to generating a calibrated alarm score, the reliability framework records entropy, expert disagreement, conformal confidence, and effective sample size as selective-risk indicators. These metrics do not influence the anomaly ranking but instead support deployment diagnostics, including abstention, risk-coverage analysis, and manual review of low-confidence alarms.

## 4. Experiments

### 4.1. Datasets and Few-Shot Splits

We conduct evaluations on MVTec AD [31] and VisA [32]; the strict external-transfer audit additionally incorporates MPDD. MVTec AD comprises industrial object and texture categories with pixel-level defect masks. VisA contains a broader range of industrial categories and serves as the primary benchmark for full-scale corruption and calibration-shift. MPDD provides six metal-part categories for an external MVTec-to-MPDD stress test; within-MPDD category certification is not pursued due to the insufficient number of categories for a three-way source split. For each class, support sets are sampled from normal training images using fixed seeds. Unless otherwise specified,  $k \in \{1, 2, 4, 8\}$  is employed for clean accuracy-storage and strict nested studies, whereas the earlier conformal corruption tables utilize  $k \in \{4, 8\}$ .

### 4.2. Evaluation Protocols

We structure the experiments into six protocol groups. The first five address clean accuracy and storage, PCA64 and PCA128 sensitivity, corruption reliability, representative routing, and attainable operating levels. The sixth protocol is the frozen strict nested protocol, which uses  $k \in \{1, 2, 4, 8\}$ , seeds 0-4,  $\alpha \in \{0.05, 0.10, 0.20\}$ , and includes clean data as well as Gaussian noise, blur, brightness/contrast, and JPEG corruptions. Each cell contains at most 120 evaluation images, with  $\rho = 0.01$ , PCA64,  $\delta = 0.05$ , and up to 20 candidate thresholds. This protocol evaluates MVTec and VisA within-dataset, MVTec-to-VisA, and MVTec-to-MPDD, using matched, clean-source, condition-agnostic, and mismatched negative-control routing. The operational gate requires a nonzero category-certified threshold in at least 80% of target cells, a target false alarm rate (FAR) of at most  $\alpha + 0.02$ , nonzero power, at least 80% no-harm, and a positive lower bound on simultaneous power gain. Zero thresholds and failed cells are retained.

### 4.3. Baselines and Ablations

We compare against a controlled nearest-neighbor memory bank utilizing DINOv2, a SubspaceAD-style PCA residual baseline, scalar Platt, Vector Platt, Shift-Aware Platt, weighted Platt, temperature scaling, isotonic regression, histogram binning, LOIO conformal, and weighted conformal. The controlled memory-bank implementation does not constitute an official reproduction of PatchCore or Anomaly-DINO; it maintains the frozen DINOv2 feature pipeline and fixed data split to isolate scoring and storage trade-offs. Additionally, the official AnomalyDINO code (ViT-S/14 at 448px, agnostic preprocessing) is executed on both MVTec and VisA (official 1-class split) for  $k \in \{1, 4, 8\}$  with three seeds, and results are reported as a separate accuracy reference. WinCLIP [10] is included as the vision-language representative, but only as reported numbers, since no official implementation is available. Auditing the leading community reimplementation under its protocol (ViT-B-16+240, LAION-400M; zero-shot and 1-shot with fixed support lists, three runs) yielded 71.7/77.4 mean image AUROC on MVTec and 66.0/70.0 on VisA, compared to the reported 91.8/93.1 and 78.1/83.8, indicating a roughly 20-point gap that the reimplementation's authors also acknowledge and cannot attribute. Consequently, claim rules distinguish audited rows from reported rows, and WinCLIP is entered in Table 2 as a cited reference rather than an audited baseline. Official SubspaceAD representative runs are included as a novelty guardrail. Their strong performance demonstrates that frozen DINOv2 subspace residuals constitute an existing competitive ranking mechanism; the residual is inherited, and the contribution of this paper commences with the reliability of the alarms constructed upon it.

### 4.4. Metrics

Ranking metrics are distinguished from reliability metrics. AUROC and AP are calculated from the raw PCA or subspace residual score  $s(x)$ . Pixel-level localization is assessed using pixel AUROC and AU-PRO when masks are available. Reliability is evaluated operationally by empirical false-alarm rate at fixed  $\alpha$ , detection power, alarm precision, discrete uniformity of normal p-values, and risk-coverage curves. Expected calibration error (ECE), Brier score, and negative log-likelihood (NLL) are reported as secondary metrics based on the selected reliability probability or probability-like conformal view. Storage is quantified as retained reference or model state per class. Robustness experiments include corruption shift and a label-aware PCA-residual FGSM surrogate, with the latter serving exclusively as a fragility diagnostic.

### 4.5. Reproducibility and Claim Rules

All experiments were conducted on a single NVIDIA RTX 5090 GPU using PyTorch 2.12.1, with DINOv2 patch features cached in fp32 format. For reproducibility, the frozen commit, DINOv2 source and weight hashes, environment specifications, dataset counts, per-image predictions, base-image identities, support manifests, corruption parameters, source partitions, candidate-threshold bounds, and

**Table 2**

Clean image-level ranking performance and category-specific storage. Controlled DINOv2 NN and PCA64/PCA128 share our feature, split, metric, and storage-accounting protocol and are averaged over classes and five seeds. The official AnomalyDINO reproduction uses three seeds under its released protocol; WinCLIP values are reported in [10]. Bold marks the best value only among common-protocol rows in each dataset/ $k$  block. “–” denotes not evaluated, N/C not directly comparable, and N/R not reported.

| Dataset | $k$ | Method               | AUROC $\uparrow$ | AP $\uparrow$ | ECE $\downarrow$ | MiB $\downarrow$ |
|---------|-----|----------------------|------------------|---------------|------------------|------------------|
| MVTec   | 1   | Official AnomalyDINO | 0.965            | 0.982         | –                | –                |
|         |     | WinCLIP (reported)   | 0.931            | N/C           | N/R              | N/R              |
|         |     | Controlled DINOv2 NN | <b>0.914</b>     | <b>0.952</b>  | 0.277            | 2.005            |
|         |     | PCA64 (ours)         | 0.904            | 0.951         | <b>0.274</b>     | <b>0.472</b>     |
|         | 4   | Official AnomalyDINO | 0.976            | 0.984         | –                | –                |
|         |     | WinCLIP (reported)   | 0.952            | N/C           | N/R              | N/R              |
|         |     | Controlled DINOv2 NN | <b>0.942</b>     | <b>0.967</b>  | 0.693            | 6.000            |
|         |     | PCA64 (ours)         | 0.937            | <b>0.967</b>  | <b>0.214</b>     | <b>0.472</b>     |
|         | 8   | Official AnomalyDINO | 0.980            | 0.990         | –                | –                |
|         |     | Controlled DINOv2 NN | <b>0.948</b>     | 0.970         | 0.693            | 6.000            |
|         |     | PCA64 (ours)         | 0.945            | <b>0.972</b>  | <b>0.154</b>     | <b>0.472</b>     |
|         |     | PCA128 (ours)        |                  |               |                  |                  |
| VisA    | 1   | Official AnomalyDINO | 0.857            | 0.866         | –                | –                |
|         |     | WinCLIP (reported)   | 0.838            | N/C           | N/R              | N/R              |
|         |     | Controlled DINOv2 NN | 0.804            | 0.809         | 0.434            | 2.005            |
|         |     | PCA64 (ours)         | 0.823            | 0.834         | <b>0.429</b>     | <b>0.472</b>     |
|         | 4   | Official AnomalyDINO | 0.912            | 0.918         | –                | –                |
|         |     | WinCLIP (reported)   | 0.873            | N/C           | N/R              | N/R              |
|         |     | Controlled DINOv2 NN | 0.862            | 0.861         | 0.543            | 6.000            |
|         |     | PCA64 (ours)         | 0.870            | 0.883         | 0.284            | <b>0.472</b>     |
|         |     | PCA128 (ours)        | <b>0.885</b>     | <b>0.895</b>  | <b>0.278</b>     | 0.566            |
|         | 8   | Official AnomalyDINO | 0.926            | 0.929         | –                | –                |
|         |     | Controlled DINOv2 NN | 0.873            | 0.870         | 0.533            | 6.000            |
|         |     | PCA64 (ours)         | 0.880            | 0.891         | 0.205            | <b>0.472</b>     |
|         |     | PCA128 (ours)        | <b>0.897</b>     | <b>0.905</b>  | <b>0.170</b>     | 0.566            |

SHA-256 checksums are documented. Automated integrity checks encompass 1,500 MVTec, 1,200 VisA, and 600 MPDD view cells, identifying duplicate cells, missing support statistics, or support-test overlap. Reliability methods cannot claim ranking improvement unless the raw score is altered; image-level and category-level certificates are not interchangeable.

## 5. Results

### 5.1. Ranking Substrate and Storage

Table 2 presents clean image-level ranking performance and category-specific storage. The official AnomalyDINO reproduction (released code, three seeds) achieves the highest image AUROC at each reported  $k$  on both datasets: 0.965, 0.976, and 0.980 on MVTec, and 0.857, 0.912, and 0.926 on VisA. Because these results follow the released method’s own evaluation protocol, they are external accuracy references rather than direct evidence of superiority under our controlled protocol. WinCLIP is likewise a cited reference: its paper reports AUROC and AUPR at 1/4 shots, but not an AP implementation verified to match ours. The reported AUPR values are 0.965/0.973 on MVTec and 0.851/0.888 on VisA; we therefore mark its AP entries N/C instead of mixing them into the AP comparison. WinCLIP does not report ECE under our calibration protocol, category-specific

storage under our accounting, or an 8-shot result. The absence of an official implementation and the community-reimplementation discrepancy are documented in Section 4.

Compared with the controlled DINOv2 nearest-neighbor ranker, the PCA64 residual ranker yields lower AUROC by 0.010/0.005/0.003 on MVTec and higher AUROC by 0.019/0.008/0.007 on VisA at  $k = 1/4/8$ , respectively, while requiring 0.472 MiB of category-specific state instead of 2.005–6.000 MiB. Accordingly, PCA64 is treated as a compact supporting ranker rather than a ranking contribution or state-of-the-art result. On VisA, PCA128 increases class-averaged AUROC over PCA64 by 0.011/0.015/0.017 at  $k = 1/4/8$ , with storage rising to 0.566 MiB; this is a descriptive accuracy-storage ablation rather than a new ranking-method claim. The controlled nearest-neighbor rows are not official PatchCore or AnomalyDINO reproductions. Boldface in Table 2 marks the best result only among methods evaluated under our common protocol, external AnomalyDINO and WinCLIP references are excluded from this comparison.

The controlled nearest-neighbor ECE values diagnose a failed scalar calibration fit rather than a failure in ranking. Under the label-free scalar-Platt protocol, exact support-patch self-matches at low  $k$  and the 4096-patch subsampling regime at larger  $k$  coincide with near-constant test probabilities. On MVTec, ECE is 0.277 at  $k = 1$  and 0.693 at



**Table 3**

Target-only LOIO operating points. The floor is the smallest p-value,  $1/(k+1)$ ; “attainable” is the largest p-value grid point not exceeding the nominal  $\alpha$ , or zero if none exists. FAR and detection are averaged over classes, seeds, and corruptions.

| dataset      | $k$ | floor | nominal $\alpha$ | attainable | FAR   | detection |
|--------------|-----|-------|------------------|------------|-------|-----------|
| VisA (full)  | 4   | 0.200 | 0.01             | 0.000      | 0.000 | 0.000     |
|              |     |       | 0.05             | 0.000      | 0.000 | 0.000     |
|              |     |       | 0.10             | 0.000      | 0.000 | 0.000     |
|              |     |       | 0.20             | 0.200      | 0.152 | 0.599     |
|              | 8   | 0.111 | 0.01             | 0.000      | 0.000 | 0.000     |
|              |     |       | 0.05             | 0.000      | 0.000 | 0.000     |
|              |     |       | 0.10             | 0.000      | 0.000 | 0.000     |
|              |     |       | 0.20             | 0.111      | 0.111 | 0.537     |
| MVTec (full) | 4   | 0.200 | 0.01             | 0.000      | 0.000 | 0.000     |
|              |     |       | 0.05             | 0.000      | 0.000 | 0.000     |
|              |     |       | 0.10             | 0.000      | 0.000 | 0.000     |
|              |     |       | 0.20             | 0.200      | 0.411 | 0.887     |
|              | 8   | 0.111 | 0.01             | 0.000      | 0.000 | 0.000     |
|              |     |       | 0.05             | 0.000      | 0.000 | 0.000     |
|              |     |       | 0.10             | 0.000      | 0.000 | 0.000     |
|              |     |       | 0.20             | 0.111      | 0.304 | 0.907     |

**Table 4**

Target-only LOIO false-alarm rate (FAR), anomaly-detection rate, and precision at the first reported attainable level,  $\alpha=0.20$ . Results are pooled over classes and seeds and reported separately by dataset, corruption, and  $k$ . Alarms fire at  $p \leq \alpha$  using the comparison tolerance described in the text.

| dataset      | corruption    | $k$ | FAR   | detection | precision |
|--------------|---------------|-----|-------|-----------|-----------|
| VisA (full)  | blur          | 4   | 0.155 | 0.610     | 0.806     |
|              |               | 8   | 0.109 | 0.546     | 0.841     |
|              | bright/contr. | 4   | 0.155 | 0.595     | 0.803     |
|              |               | 8   | 0.110 | 0.533     | 0.836     |
|              | Gauss. noise  | 4   | 0.142 | 0.585     | 0.813     |
|              |               | 8   | 0.103 | 0.522     | 0.843     |
|              | JPEG          | 4   | 0.157 | 0.604     | 0.803     |
|              |               | 8   | 0.123 | 0.547     | 0.825     |
| MVTec (full) | blur          | 4   | 0.306 | 0.887     | 0.872     |
|              |               | 8   | 0.194 | 0.908     | 0.916     |
|              | bright/contr. | 4   | 0.313 | 0.887     | 0.869     |
|              |               | 8   | 0.188 | 0.908     | 0.919     |
|              | Gauss. noise  | 4   | 0.463 | 0.879     | 0.816     |
|              |               | 8   | 0.355 | 0.898     | 0.855     |
|              | JPEG          | 4   | 0.410 | 0.890     | 0.836     |
|              |               | 8   | 0.307 | 0.914     | 0.875     |

$k = 4, 8$ , which correspond to the class-averaged prevalence complements; the Brier score is similarly close to ECE. These degenerate probabilities are not used as a competitive calibration baseline.

*Storage and compute accounting.* Storage is reported in MiB and includes only category-specific state, excluding the shared frozen DINOv2 ViT-S/14 backbone ( $\approx 22$ M parameters,  $\approx 84$  MiB in fp32), which is stored once per deployment for all compared frozen-feature methods. The 0.472 MiB PCA64-based reliability state comprises the PCA mean and 64-component basis ( $\approx 0.095$  MiB), the synthetic-anomaly MLP head, and its feature-normalization statistics. Expanding the basis from 64 to 128 components adds

0.094 MiB. The fitted three-feature Vector Platt scaler requires only tens of bytes, which is below the displayed precision. The controlled memory bank stores up to 4096 fp32 vectors of dimension 384, totaling exactly 6.000 MiB at maximum capacity. For cached features, raw PCA-residual scoring requires approximately 1.3 ms per image, compared with 5 to 13 ms for controlled nearest-neighbor scoring. These timings exclude shared DINOv2 feature extraction and reflect only category-specific raw-score computation. LOIO also requires a one-time setup of  $k$  leave-one-out PCA fits per category, each using up to  $(k - 1) \times 1369$  support patches.

**Table 5**

Frozen strict nested CRESS audit over four routing modes,  $k \in \{1, 2, 4, 8\}$ , three  $\alpha$  levels, five seeds, and five conditions. “Min UCB” is the smallest observed candidate upper bound. Nonzero is the fraction of target cells receiving  $\tau > 0$ . Category and image rows target different estimands.

| source $\rightarrow$ target | certification unit  | units/cell | min UCB | nonzero | interpretation          |
|-----------------------------|---------------------|------------|---------|---------|-------------------------|
| MVTec $\rightarrow$ MVTec   | category            | 4          | 0.950   | 0.000   | new-category gate fails |
| VisA $\rightarrow$ VisA     | category            | 3          | 1.000   | 0.000   | new-category gate fails |
| MVTec $\rightarrow$ VisA    | category            | 4          | 0.961   | 0.000   | new-category gate fails |
| MVTec $\rightarrow$ MPDD    | category            | 4          | 0.986   | 0.000   | new-category gate fails |
| MVTec $\rightarrow$ MVTec   | fixed-archive image | 75–191     | 0.039   | 0.367   | sensitivity only        |
| VisA $\rightarrow$ VisA     | fixed-archive image | 150–180    | 0.042   | 0.604   | sensitivity only        |
| MVTec $\rightarrow$ VisA    | fixed-archive image | 72–159     | 0.048   | 0.415   | sensitivity only        |
| MVTec $\rightarrow$ MPDD    | fixed-archive image | 81–190     | 0.040   | 0.370   | sensitivity only        |

**Table 6**

Full VisA conformal reliability under four corruptions, pooled over classes and seeds. The PCA/subspace ranking score is fixed, so AUROC and AP do not change between reliability routes;  $N$  is the number of evaluated corruption-view rows. Bold marks the lower ECE, Brier score, and NLL within each group.

| Group | Reliability route  | N     | AUROC | AP    | ECE $\downarrow$ | Brier $\downarrow$ | NLL $\downarrow$ |
|-------|--------------------|-------|-------|-------|------------------|--------------------|------------------|
| All   | LOIO conformal     | 56000 | 0.823 | 0.846 | <b>0.077</b>     | <b>0.187</b>       | <b>0.686</b>     |
|       | Weighted conformal | 56000 | 0.823 | 0.846 | 0.166            | 0.244              | 0.813            |
| $k=4$ | LOIO conformal     | 28000 | 0.820 | 0.842 | <b>0.039</b>     | <b>0.187</b>       | <b>0.760</b>     |
|       | Weighted conformal | 28000 | 0.820 | 0.842 | 0.216            | 0.259              | 0.925            |
| $k=8$ | LOIO conformal     | 28000 | 0.828 | 0.851 | <b>0.114</b>     | <b>0.188</b>       | <b>0.612</b>     |
|       | Weighted conformal | 28000 | 0.828 | 0.851 | 0.116            | 0.229              | 0.701            |

The clean benchmark applies max pooling, whereas the conformal pipeline uses the top-1% mean. The representative aggregation ablation in Table 15 (Appendix A) demonstrates that both choices are within 0.01 to 0.02 AUROC of the best tested setting. These results indicate that pooling does not drive the ranking comparison, but it does not establish invariance outside the tested grid.

## 5.2. Target-Only Resolution and False-Alarm Behavior

Calibration error does not address the operator’s main concern: “if an alarm fires at  $p \leq \alpha$ , how often is a normal part flagged?” Table 3 first isolates the structural resolution limit. With  $k = 4$ , the smallest p-value is  $1/(k + 1) = 0.2$ , so nominal levels 0.01, 0.05, and 0.10 cannot raise an alarm on either dataset. With  $k = 8$ , the floor decreases to  $1/9 \approx 0.111$ , but the same three tested levels remain below it. Their zero FAR and detection values are therefore resolution artifacts of the finite calibration set, not evidence of effective conservative control.

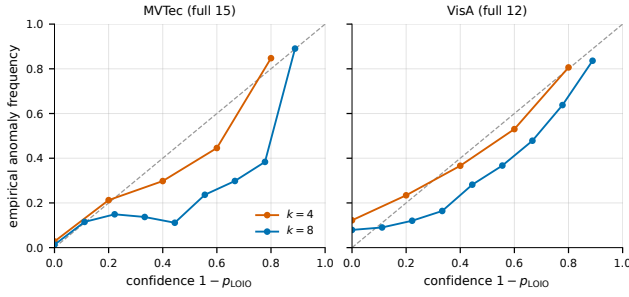
At nominal  $\alpha = 0.20$ , the effective attainable threshold is 0.20 for  $k = 4$  but only  $1/9$  for  $k = 8$ , because no  $k = 8$  grid point lies between  $1/9$  and 0.20. On VisA, the resulting aggregate FAR/detection pairs are 0.152/0.599 at  $k = 4$  and 0.111/0.537 at  $k = 8$ ; on MVTec they are 0.411/0.887 and 0.304/0.907, respectively. Thus the same nominal  $\alpha$  can induce different effective p-value thresholds across  $k$ , and the observed FAR itself is not restricted to the p-value grid. The aggregate contrast already indicates dataset-dependent

behavior: VisA remains below nominal, whereas MVTec exceeds it substantially.

Table 4 resolves the  $\alpha = 0.20$  aggregate by corruption. For VisA, FAR remains below the nominal level across all four corruptions, ranging from 0.142 to 0.157 at  $k = 4$  and from 0.103 to 0.123 at  $k = 8$ . These operating points maintain nonzero detection rates of 0.585 to 0.610 and 0.522 to 0.547, respectively, with precision values between 0.803 and 0.843 under the benchmark prevalence.

In contrast, MVTec exhibits less stable behavior. At  $k = 4$ , FAR ranges from 0.306 to 0.463 across corruptions, despite detection rates of 0.879 to 0.890. Increasing  $k$  to 8 reduces FAR in every corruption cell and increases observed precision, but does not consistently restore nominal behavior. Specifically, blur and brightness/contrast are close to  $\alpha$  (0.194 and 0.188), while Gaussian noise and JPEG remain anti-conservative (0.355 and 0.307). Thus, crossing the resolution floor results in nonzero alarms but does not guarantee stable false-alarm behavior under corruption shift. Increasing  $k$  mitigates, but does not eliminate, the observed mismatch. Precision is reported descriptively, as it depends on the anomaly prevalence in the evaluation set.

A relevant numerical implementation detail is that an fp32 representation of  $1/5$  may be 0.20000000298. Therefore, exact comparison with 0.2 can inadvertently exclude all floor-level alarms. All reported thresholding procedures apply a small comparison tolerance.



**Figure 2:** LOIO reliability diagrams on the full MVTEC (left) and VisA (right) corruption benchmarks, pooled over classes, seeds, and corruptions. Points occupy the  $k+1$  attainable confidence levels; positions below the diagonal indicate over-confidence.

### 5.3. Strict CRESS Category Certification

Table 5 summarizes the frozen strict nested audit. The primary category-level result is uniformly negative. The 960 gate configurations include four source–target jobs, four routing modes, four  $k$  values, three  $\alpha$  levels, and five conditions. Each configuration is evaluated across the declared target cells and five support seeds when computing its nonzero-threshold rate. None achieves the required 80% rate, as every category-certified target cell receives  $\tau = 0$ . The category block applies only four certification categories per cell for MVTEC and both transfer directions, and three for VisA. The smallest observed candidate UCB is 0.950 on MVTEC, 1.000 on VisA, 0.961 for MVTEC-to-VisA, and 0.986 for MVTEC-to-MPDD. Since all values exceed the largest tested  $\alpha = 0.20$ , no positive candidate passes the category-risk gate.

This result does not indicate a GPU or optimization failure. Proposition 3 demonstrates that with  $A = 3$ ,  $\delta = 0.05$ , and even  $M = 1$ , a zero-loss Hoeffding gate needs at least 60 categories at  $\alpha = 0.20$ . Proposition 4 further indicates that this outcome is not solely a consequence of Hoeffding looseness: even for one fixed candidate without a multiplicity penalty, any deterministic, uniformly valid, distribution-free 95% upper bound based only on iid category losses needs at least 14 all-zero categories to reach 0.20; the frozen family allocation increases this requirement to 22. Both thresholds exceed the available three or four independent certification categories.

The lower block of Table 5 shifts the certification unit from categories to images within a fixed source archive. With 72 to 191 image units per cell, the smallest observed UCB decreases to 0.039 to 0.048, which is below at least one tested  $\alpha$ . Positive thresholds are observed in 36.7% of MVTEC, 60.4% of VisA, 41.5% of MVTEC-to-VisA, and 37.0% of MVTEC-to-MPDD cells. However, none of these aggregate rates meet the 80% operational criterion. More significantly, the increased image count tightens the bound for the fixed archive mixture but does not introduce independent categories or establish reliability for a new target

category. Repeated corruption views are excluded as independent images. Thus, the image-level result serves as a sensitivity analysis rather than a resolution of the category-level failure or evidence of target-category false-alarm control.

### 5.4. Conformal Reliability under Corruption

Table 6 reports the full VisA aggregate results using a fixed PCA/subspace ranking score. Across 56,000 corruption-view rows, LOIO conformal achieves a pooled ECE of 0.0766, with values of 0.0391 at  $k = 4$  and 0.1140 at  $k = 8$ . Weighted conformal exhibits a substantially higher ECE at  $k = 4$  and converges toward LOIO at  $k = 8$  (0.116 versus 0.114). The near-tie in  $k = 8$  ECE does not extend to the other reported losses: LOIO has lower Brier score (0.188 versus 0.229) and NLL (0.612 versus 0.701). AUROC and AP remain identical between the two routes in every group because the underlying anomaly score is fixed; thus, the reliability transformation does not claim to improve ranking.

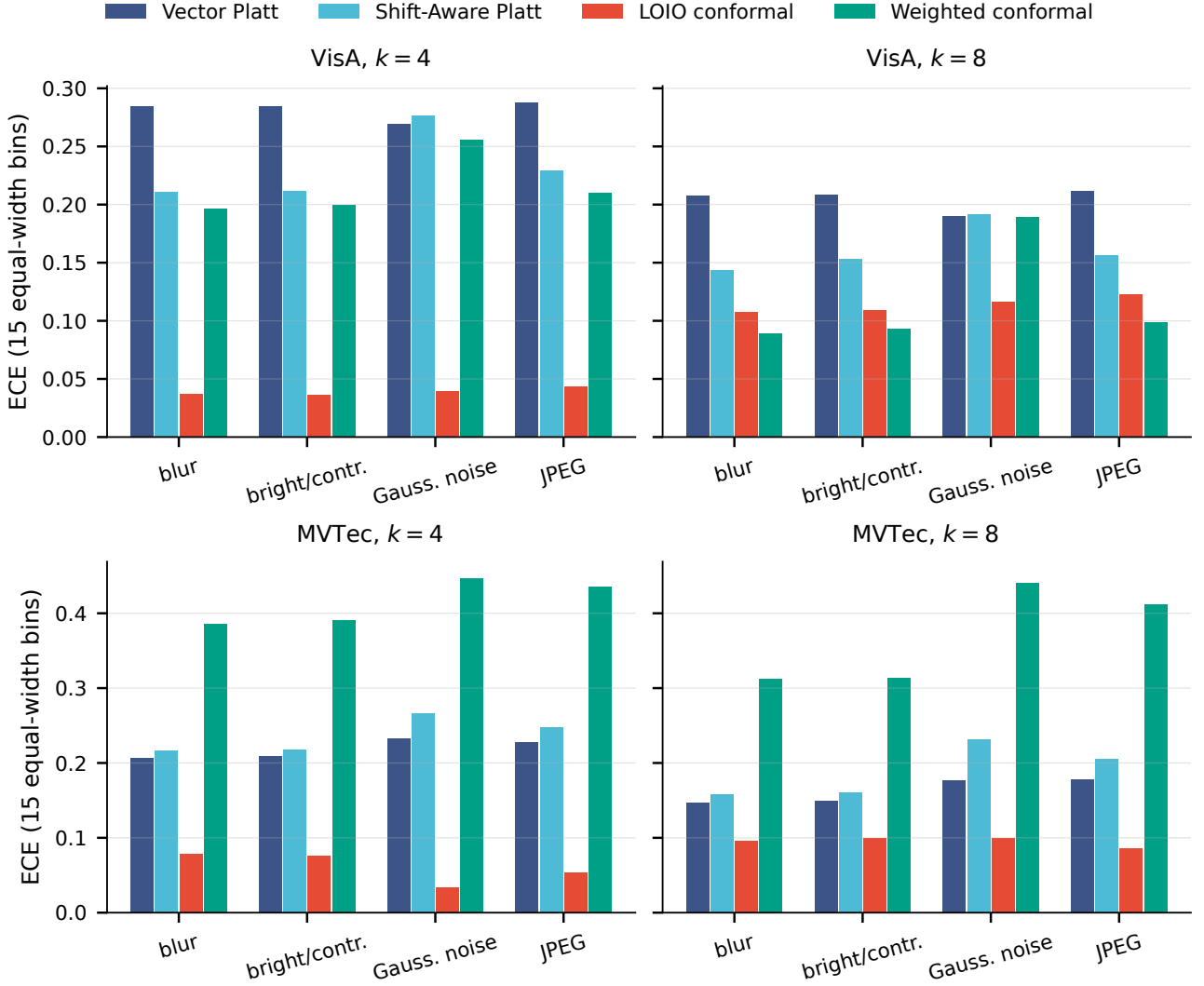
In the comprehensive MVTEC study (15 classes,  $k \in \{4, 8\}$ , seeds 0–2, four corruptions, 37,464 rows), LOIO achieves an overall ECE of 0.0684 (0.0600 at  $k = 4$  and 0.0769 at  $k = 8$ ) while maintaining an image AUROC of 0.912. LOIO consistently yields lower ECE than both Vector Platt and Shift-Aware Platt across all eight  $k$ –corruption settings; for instance, at MVTEC  $k = 4$  under Gaussian noise, ECE is 0.033 compared to 0.233 and 0.270, respectively. This pattern is descriptive across the tested cells and does not represent a confirmatory significance result. ECE remains a secondary metric and does not guarantee control over false alarms.

Table 7 (Appendix A) presents VisA per-corruption values, and Figure 3 compares both datasets. LOIO has the lowest ECE in all four VisA  $k = 4$  cells and under Gaussian noise at  $k = 8$ , whereas weighted conformal achieves lower ECE for the remaining three VisA  $k = 8$  corruptions. Across the MVTEC panels, LOIO consistently exhibits the lowest ECE in every displayed cell. This cellwise summary provides a more accurate characterization than describing either route as uniformly superior.

Figure 2 reveals the distributional patterns that scalar ECE summaries obscure. The attainable confidence values are sparse because the LOIO construction yields only  $k + 1$  p-value levels. In both datasets, the pooled  $k = 8$  curve lies further below the diagonal than the  $k = 4$  curve in the upper-confidence region, which aligns with the higher pooled LOIO ECE at  $k = 8$ . As these diagrams aggregate across classes, seeds, and corruptions, they diagnose overall miscalibration but do not indicate that every category exhibits the same direction or magnitude of error.

### 5.5. Distributional and Calibration Diagnostics

Conformal p-values satisfy  $P(p \leq \alpha) \leq \alpha$  for normal data only when exchangeability conditions are met. This property reflects a false-alarm rate rather than a posterior probability. Because a  $k$ -shot LOIO p-value is confined to the grid  $\{j/(k + 1)\}$ , the application of a continuous Kolmogorov–Smirnov test against  $U(0, 1)$  is inappropriate.



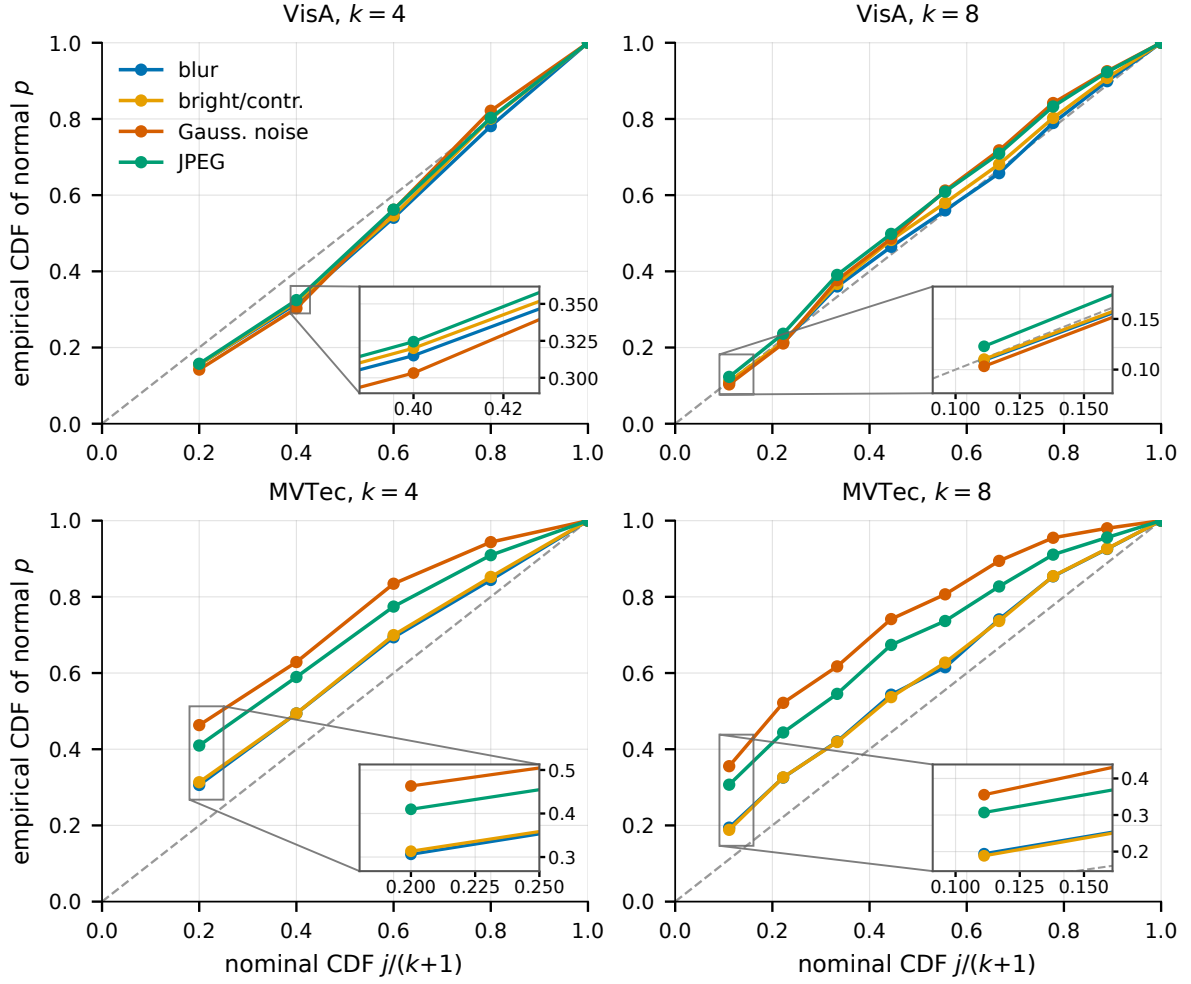
**Figure 3:** ECE by corruption and reliability route on full VisA (top) and full MVTec (bottom) at  $k=4$  and  $k=8$ . Lower values indicate better calibration under the reported ECE definition.

Figure 4 instead compares the empirical cumulative distribution function (CDF) of normal p-values to the ideal discrete-uniform reference at each grid point. Deviations are conservative for VisA at  $k = 4$  (with the largest negative gap of  $-0.10$  at  $0.4$ ), whereas VisA at  $k = 8$  and MVTec at both  $k$  values become anti-conservative under corruption. For instance, for MVTec  $k = 4$  with Gaussian noise,  $\hat{F}(0.2) = 0.46$  compared to the reference value of  $0.20$ . Because rows share calibration sets, classes, and repeated seeds, a cluster-preserving exchangeable Monte Carlo diagnostic is employed rather than an independent and identically distributed (iid) confirmatory test. The results indicate that corrupted normal residuals increase relative to clean LOIO calibration scores, although a causal mechanism is not identified. A matched-LOIO audit on representative MVTec  $k = 4$  classes increases the clean FAR from  $0.17$  to  $0.25$ , with power remaining essentially unchanged. Thus,

fold asymmetry alone does not account for the observed behavior.

Table 8 presents comparisons across class, seed, and corruption for Vector Platt, Shift-Aware Platt, temperature scaling, isotonic regression, histogram binning, and scalar Platt, all evaluated under a consistent label-free protocol. In contrast to the pooled expected calibration error (ECE) in Table 6, these results first compute ECE within each class–seed–corruption cell and then report the mean and standard deviation across cells; as a result, the two summaries are not necessarily numerically equivalent. LOIO demonstrates lower mean per-cell ECE in most comparisons (for example, MVTec  $k = 4$ : LOIO  $0.120 \pm 0.054$  versus isotonic  $0.222 \pm 0.088$ , temperature  $0.240 \pm 0.104$ , and histogram binning  $0.229 \pm 0.099$ ), although the margin decreases at  $k = 8$ . LOIO achieves lower ECE than isotonic regression in 55% of MVTec cells and 62% of VisA cells. Against Shift-Aware Platt on VisA  $k = 8$ , the





**Figure 4:** Empirical CDF of normal-image LOIO p-values at each attainable point  $j/(k+1)$  versus the idealized exchangeable reference (dashed diagonal), by corruption. Points above the diagonal indicate excess false alarms relative to the reference; lower-right insets enlarge the first informative grid region.

mean difference is  $-0.011$ , with LOIO exhibiting lower ECE in 51% of cells. Historical unadjusted significance values are not reported because a complete multiplicity-adjusted analysis is unavailable, and no confirmatory significance claims are made. Equal-mass ECE increases LOIO’s per-cell error compared with equal-width ECE, underscoring the importance of the operational analyses discussed above.

ECE on  $1 - p$  is sensitive to anomaly prevalence, even when the ranking remains unchanged. On the full VisA dataset, LOIO ECE decreases from 0.404 at 1% anomaly prevalence to 0.149 at 50%, and the ordering between LOIO and weighted methods reverses across this range. Consequently, anomaly-balanced ECE does not guarantee deployment reliability. Throughout this paper, FAR, power, precision, and risk-coverage are used for operational interpretation.

## 5.6. Historical, Non-Independent CRESS Sensitivity

These analyses were performed before implementing the strict nested protocol and are presented exclusively as empirical sensitivity checks. As they reuse source evidence and employ pointwise rather than simultaneous intervals, these results do not supersede the category-level findings reported in Section 5.3.

Across all 15 label-stratified MVTec classes at  $k = 4$  (seeds 0–2, five conditions, 675 evaluation cells), matched-condition CRESS demonstrates mean power of 0.216 and 0.416 at  $\alpha = 0.05$  and 0.10, respectively, when the target-only anchor is inactive. The mean FAR is 0.050, 0.105, and 0.216 at  $\alpha = 0.05$ , 0.10, and 0.20, respectively. Precision exceeds 0.90, and no-harm rates are 89%, 82%, and 89%. Although the historical empirical gate generally passes, certain condition-level exceptions remain: JPEG FAR is 0.121 at  $\alpha = 0.10$ , and Gaussian noise and JPEG reach 0.226 and 0.235 at  $\alpha = 0.20$ , surpassing the  $\alpha + 0.02$  thresholds.

Table 9 further illustrates the power and FAR trade-off when the target-only anchor is active. At  $\alpha = 0.20$ , historical CRESS reduces FAR compared to the anchor in every condition by 0.058 to 0.234, but its power is also reduced by 0.094 to 0.365. The most substantial simultaneous reduction in FAR and power occurs under Gaussian noise. These paired changes indicate an empirical threshold-selection trade-off rather than superiority of CRESS over the anchor.

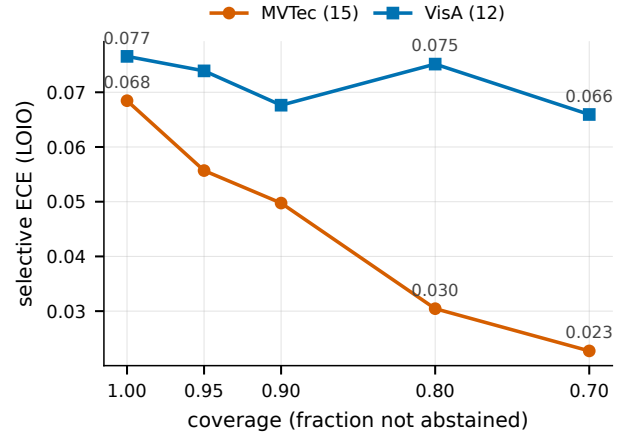
Supporting tables are available in Appendix A. At  $k = 8$ , matched-condition CRESS achieves pooled power of 0.547 and 0.693 at the two nominal levels, both below the target-only floor. The pooled FARs of 0.052, 0.121, and 0.230, as well as no-harm rates of 80%, 74%, and 81%, meet the historical empirical budget only at  $\alpha = 0.05$  (Table 10). The breakdown by condition is less uniform than the pooled values. At  $\alpha = 0.10$ , clean, blur, and brightness/contrast remain within the  $\alpha + 0.02$  budget, while Gaussian noise and JPEG reach FAR values of 0.138 and 0.141, respectively. At  $\alpha = 0.20$ , only the clean condition remains within budget; blur is marginally above at 0.221, and brightness/contrast, Gaussian noise, and JPEG reach 0.223, 0.232, and 0.260, respectively. Despite these results, sub-floor precision remains high (0.936 to 0.976), indicating that the primary limitation is in false-alarm control rather than a lack of detected anomalies.

Smoothed randomized p-values enable alarms below the deterministic floor but maintain the target-only mismatch. On pooled MVTEC at  $\alpha = 0.05$  and 0.10, randomization yields FAR values of 0.085 and 0.170 and power of 0.216 and 0.431, respectively, whereas historical CRESS yields FAR values of 0.049 and 0.105 and power of 0.216 and 0.416 (Table 11). On VisA, randomization is more conservative, with FAR values of 0.036 and 0.072 compared to 0.051 and 0.095. Its power is slightly higher at  $\alpha = 0.05$  (0.150 versus 0.144) but lower at  $\alpha = 0.10$  (0.300 versus 0.351). Therefore, neither approach consistently outperforms the other across datasets and operating levels. The historical CRESS comparison remains non-independent and cannot be considered certified evidence.

Within-VisA sources yield FAR values of 0.051, 0.095, and 0.192 and power of 0.144, 0.351, and 0.637, with precision of 0.847, 0.843, and 0.810 and no-harm rates of 89%, 84%, and 78% (Table 12). Replacing these with MVTEC sources reduces FAR to 0.013, 0.040, and 0.098 but also decreases power to 0.091, 0.201, and 0.415. Precision increases to 0.936, 0.881, and 0.852, and no-harm rates rise to 97%, 93%, and 93%. The foreign archive is therefore more conservative in this transfer direction; however, the associated reduction in power precludes interpreting this as a general safety property.

## 5.7. Secondary Deployment and Detector Diagnostics

The risk-coverage analysis presented in Figure 5 ranks images according to predictive entropy. Abstaining from the most uncertain 20% reduces LOIO ECE from 0.068 to 0.030 on the full MVTEC dataset, but only from 0.077



**Figure 5:** LOIO risk-coverage curves with entropy-ranked abstention on full MVTEC and VisA. Coverage is the retained image fraction; risk is ECE on the retained subset.

to 0.075 on VisA. Notably, VisA achieves 0.066 at 70% coverage. These results indicate that the benefit of abstention is dataset-dependent rather than a uniform outcome. This analysis serves as a selective-operation diagnostic rather than a formal certificate.

Table 13 evaluates protocol-locked representative routing. Employing fixed LOIO reduces ECE from 0.301–0.361 to 0.099–0.154 across both transfer directions, LOCO, and within-split evaluations. The corresponding  $\Delta$ ECE ranges from  $-0.207$  to  $-0.188$ , with 100% no-harm observed in the reported cells. These findings support using a simple fixed target-only route as the main baseline, without validation-label expert selection. Because the table covers representative locked protocols and ECE only, it does not demonstrate that LOIO outperforms learned or weighted routing across all datasets, shifts, or operational metrics.

Table 14 distinguishes between localization and robustness. PCA64 surpasses the controlled memory bank in pixel AUROC by only 0.004, 0.006, and 0.008 at  $k = 1$ ,  $k = 4$ , and  $k = 8$ , respectively. AU-PRO reveals a complementary trade-off: PCA64 performs lower at  $k = 1$  (0.820 versus 0.845), matches at  $k = 4$  (0.862), and outperforms at  $k = 8$  (0.876 versus 0.863). Thus, the low-storage ranker remains competitive in this controlled localization setting, but the small, metric-dependent differences do not justify a broad claim of localization superiority.

The fragility block yields less favorable results. A label-aware PCA-residual FGSM surrogate reduces image AUROC from 0.937 and 0.945 to 0.158 and 0.175 at  $\epsilon = 2/255$ , which is below chance due to inverted score ordering. The degradation is non-monotonic; at  $8/255$ , AUROC recovers to 0.446 for both  $k$  values. This pattern demonstrates sensitivity to the surrogate objective and attack configuration. These findings are diagnostic and do not substantiate an adversarial-robustness claim.

## 6. Limitations

The evidence is limited to MVTec, VisA, and MPDD. The frozen DINOv2 PCA residual is an inherited ranking substrate, while official AnomalyDINO and reported WinCLIP results use different protocols and serve only as external references. The controlled shift study covers four synthetic corruptions, not the full range of temporal drift, sensor replacement, process changes, correlated failures, or deployment prevalence. Moreover, scoring latency is measured on cached DINOv2 features and excludes backbone inference, preprocessing, and I/O. Generalization to prospective factory operation therefore requires evaluation under real shifts and end-to-end deployment constraints.

Target-only conformal validity requires representative normal supports and the relevant exchangeability conditions. LOIO calibration and full-support test residuals are asymmetric, and their p-values occupy a finite grid; our clustered discrete-reference analyses diagnose departures from the exchangeable pattern but do not restore validity when exchangeability fails. ECE is additionally sensitive to prevalence and binning, so it remains descriptive rather than a false-alarm guarantee. The FGSM results define a separate boundary: robustness to the evaluated corruptions does not imply adversarial robustness.

The certification claims are narrower still. Historical CRESS results reuse source evidence and are empirical diagnostics, whereas the independent strict evaluation returns  $\tau = 0$  for every category-certified target cell because only three or four certification categories are available. The distribution-free all-zero lower bound explains why additional images or seeds cannot repair this category shortage; assumption-dependent parametric or hierarchical alternatives would require their own validity arguments. Fixed-archive image certificates target an archive mixture, not a new-category risk. Finally, matched routing assumes reliable condition metadata, and source-to-target control still requires source-risk dominance or category exchangeability. Thus the current evidence supports an auditable source-assisted procedure and its feasibility boundary, not a deployable certified alarm for an arbitrary target category.

## 7. Conclusion and future work

We presented an auditable separation between low-storage anomaly ranking and few-shot alarm reliability. Three conclusions follow. First, a target-only conformal alarm built from  $k$  calibration scores cannot fire below  $\alpha = 1/(k+1)$ ; zero alarms below this floor are a resolution artifact, not evidence of false-alarm control. Second, crossing the floor does not guarantee stability: corrupted normal inputs can make empirical false-alarm behavior anti-conservative without any change to the underlying ranker. Third, source-assisted category certification is limited by the number of independent categories, not by the number of images or support seeds. In the strict CRESS evaluation, every category-certified target cell receives  $\tau = 0$ , and all 960 gate configurations fail across MVTec, VisA, and two

transfer directions. This is not merely Hoeffding looseness: even without multiplicity, a deterministic distribution-free 95% category-risk upper bound requires at least 14 all-zero categories to reach  $\alpha = 0.20$ , compared with the three or four available. Fixed-archive image certificates and historical CRESS results provide sensitivity evidence for different or non-independent estimands; neither establishes certified target-category control.

The practical contribution is therefore an audit discipline rather than an unconditional deployment guarantee: report attainable rather than nominal operating points, test normal false alarms under declared shifts, specify the certification unit before observing results, and count independent categories rather than repeated views. CRESS remains a source-assisted thresholding and certification procedure, but its category-level certificate is infeasible at the archive sizes studied here. Future work must either collect category-rich source archives or introduce pre-specified parametric, hierarchical, or side-information assumptions with independent validity checks; a nominally tighter distribution-free bound cannot overcome the all-zero category limit with only three or four categories. Prospective evaluation should additionally cover real production shifts, temporal dependence, condition-estimation error, and end-to-end deployment cost.

## CRedit authorship contribution statement

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

The datasets analyzed during the current study are publicly available: MVTec AD (<https://www.mvtec.com/company/research/datasets/mvtec-ad>) and VisA (<https://github.com/amazon-science/spot-diff>). The final review package will include derived per-image prediction tables, evaluation summaries, sampling and source-partition manifests, and check-sums; the reviewer-accessible repository identifier must be inserted before submission.

### Code availability

The experimental code includes the conformal protocols, calibrators, corruption pipeline, artifact audits, and table/figure generators. A reviewer-accessible repository and immutable commit identifier must be inserted here before submission.

### Declaration of generative AI and AI-assisted technologies

During manuscript preparation, the authors used OpenAI Codex for code review, test generation, statistical-claim auditing, and language editing. No experimental observation

was generated or altered by the tool. The authors reviewed the resulting code and text and take responsibility for the content.

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## A. Supplementary tables

This appendix collects supporting tables summarized in the main text: the per-corruption VisA ECE breakdown and standard-calibrator comparison (Tables 7 and 8); the historical, non-independent CRESS sensitivity studies (Tables 9–12); protocol-routing, localization, and fragility diagnostics (Tables 13 and 14); the aggregation-sensitivity ablation (Table 15); and the novelty-positioning summary (Table 16).

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**Table 7**

ECE by reliability route on the full VisA corruption benchmark, pooled over classes and seeds and reported separately for each  $k$ -corruption cell. Lower is better, and bold marks the lowest ECE in each cell. Figure 3 gives the corresponding visual comparison for VisA and MVTEC.

| k | Corruption          | Vector | Shift-aware | LOIO         | Weighted     |
|---|---------------------|--------|-------------|--------------|--------------|
| 4 | blur                | 0.284  | 0.211       | <b>0.037</b> | 0.197        |
|   | brightness/contrast | 0.285  | 0.212       | <b>0.036</b> | 0.199        |
|   | gaussian noise      | 0.270  | 0.276       | <b>0.040</b> | 0.256        |
|   | jpeg                | 0.288  | 0.230       | <b>0.043</b> | 0.210        |
| 8 | blur                | 0.208  | 0.144       | 0.108        | <b>0.089</b> |
|   | brightness/contrast | 0.209  | 0.153       | 0.109        | <b>0.093</b> |
|   | gaussian noise      | 0.190  | 0.191       | <b>0.117</b> | 0.189        |
|   | jpeg                | 0.212  | 0.156       | 0.123        | <b>0.099</b> |

**Table 8**

LOIO conformal reliability versus label-free calibrators on the historical full corruption benchmarks. Entries are per-cell ECE mean $\pm$ standard deviation;  $\Delta$  is LOIO minus baseline (negative favors LOIO), and “LOIO better” is its per-cell win fraction.

| dataset / k | baseline            | base ECE          | LOIO ECE                          | $\Delta$ | LOIO better |
|-------------|---------------------|-------------------|-----------------------------------|----------|-------------|
| MVTEC $k=4$ | Vector Platt        | 0.223 $\pm$ 0.095 | <b>0.120<math>\pm</math>0.054</b> | -0.104   | 84%         |
|             | Shift-Aware Platt   | 0.241 $\pm$ 0.104 | <b>0.120<math>\pm</math>0.054</b> | -0.121   | 88%         |
|             | Temperature scaling | 0.240 $\pm$ 0.104 | <b>0.120<math>\pm</math>0.054</b> | -0.120   | 87%         |
|             | Isotonic regression | 0.222 $\pm$ 0.088 | <b>0.120<math>\pm</math>0.054</b> | -0.103   | 87%         |
|             | Histogram binning   | 0.229 $\pm$ 0.099 | <b>0.120<math>\pm</math>0.054</b> | -0.110   | 84%         |
|             | Scalar Platt        | 0.268 $\pm$ 0.127 | <b>0.120<math>\pm</math>0.054</b> | -0.148   | 93%         |
| MVTEC $k=8$ | Vector Platt        | 0.167 $\pm$ 0.065 | <b>0.135<math>\pm</math>0.055</b> | -0.033   | 66%         |
|             | Shift-Aware Platt   | 0.192 $\pm$ 0.088 | <b>0.135<math>\pm</math>0.055</b> | -0.057   | 75%         |
|             | Temperature scaling | 0.177 $\pm$ 0.080 | <b>0.135<math>\pm</math>0.055</b> | -0.042   | 69%         |
|             | Isotonic regression | 0.157 $\pm$ 0.069 | <b>0.135<math>\pm</math>0.055</b> | -0.023   | 55%         |
|             | Histogram binning   | 0.162 $\pm$ 0.080 | <b>0.135<math>\pm</math>0.055</b> | -0.027   | 61%         |
|             | Scalar Platt        | 0.256 $\pm$ 0.132 | <b>0.135<math>\pm</math>0.055</b> | -0.121   | 91%         |
| VisA $k=4$  | Vector Platt        | 0.286 $\pm$ 0.060 | <b>0.131<math>\pm</math>0.063</b> | -0.155   | 97%         |
|             | Shift-Aware Platt   | 0.237 $\pm$ 0.087 | <b>0.131<math>\pm</math>0.063</b> | -0.106   | 90%         |
|             | Temperature scaling | 0.312 $\pm$ 0.085 | <b>0.132<math>\pm</math>0.059</b> | -0.180   | 98%         |
|             | Isotonic regression | 0.239 $\pm$ 0.073 | <b>0.132<math>\pm</math>0.059</b> | -0.106   | 90%         |
|             | Histogram binning   | 0.301 $\pm$ 0.087 | <b>0.132<math>\pm</math>0.059</b> | -0.168   | 96%         |
|             | Scalar Platt        | 0.304 $\pm$ 0.145 | <b>0.132<math>\pm</math>0.059</b> | -0.171   | 90%         |
| VisA $k=8$  | Vector Platt        | 0.199 $\pm$ 0.063 | <b>0.151<math>\pm</math>0.054</b> | -0.048   | 81%         |
|             | Shift-Aware Platt   | 0.162 $\pm$ 0.075 | <b>0.151<math>\pm</math>0.054</b> | -0.011   | 51%         |
|             | Temperature scaling | 0.247 $\pm$ 0.089 | <b>0.158<math>\pm</math>0.051</b> | -0.089   | 75%         |
|             | Isotonic regression | 0.194 $\pm$ 0.068 | <b>0.158<math>\pm</math>0.051</b> | -0.035   | 62%         |
|             | Histogram binning   | 0.242 $\pm$ 0.097 | <b>0.158<math>\pm</math>0.051</b> | -0.083   | 69%         |
|             | Scalar Platt        | 0.418 $\pm$ 0.100 | <b>0.158<math>\pm</math>0.051</b> | -0.259   | 98%         |

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**Table 9**

Historical matched-condition CRESS versus the target-only LOIO anchor on 15 label-stratified MVTec classes ( $k=4$ , seeds 0–2). Threshold selection and source assessment are not independent. Bold marks cells with nonzero power and  $\text{FAR} \leq \alpha+0.02$ .

| corruption    | $\alpha$ | CRESS (historical) |              |       | Target-only anchor |       |       |
|---------------|----------|--------------------|--------------|-------|--------------------|-------|-------|
|               |          | FAR                | power        | prec. | FAR                | power | prec. |
| clean         | 0.05     | <b>0.050</b>       | <b>0.267</b> | 0.928 | 0.000              | 0.000 | –     |
|               | 0.10     | <b>0.100</b>       | <b>0.470</b> | 0.957 | 0.000              | 0.000 | –     |
|               | 0.20     | <b>0.209</b>       | <b>0.768</b> | 0.914 | 0.267              | 0.863 | 0.898 |
| blur          | 0.05     | <b>0.048</b>       | <b>0.242</b> | 0.915 | 0.000              | 0.000 | –     |
|               | 0.10     | <b>0.099</b>       | <b>0.468</b> | 0.950 | 0.000              | 0.000 | –     |
|               | 0.20     | <b>0.207</b>       | <b>0.760</b> | 0.904 | 0.269              | 0.860 | 0.895 |
| bright/contr. | 0.05     | <b>0.050</b>       | <b>0.254</b> | 0.914 | 0.000              | 0.000 | –     |
|               | 0.10     | <b>0.101</b>       | <b>0.467</b> | 0.954 | 0.000              | 0.000 | –     |
|               | 0.20     | <b>0.205</b>       | <b>0.769</b> | 0.905 | 0.268              | 0.863 | 0.898 |
| Gauss. noise  | 0.05     | <b>0.048</b>       | <b>0.184</b> | 0.934 | 0.000              | 0.000 | –     |
|               | 0.10     | <b>0.105</b>       | <b>0.295</b> | 0.929 | 0.000              | 0.000 | –     |
|               | 0.20     | 0.226              | 0.495        | 0.914 | 0.460              | 0.860 | 0.844 |
| JPEG          | 0.05     | <b>0.051</b>       | <b>0.134</b> | 0.911 | 0.000              | 0.000 | –     |
|               | 0.10     | 0.121              | 0.381        | 0.936 | 0.000              | 0.000 | –     |
|               | 0.20     | 0.235              | 0.633        | 0.917 | 0.434              | 0.867 | 0.854 |

**Table 10**

Historical matched-condition CRESS versus the target-only LOIO anchor on 15 label-stratified MVTec classes ( $k=8$ , seeds 0–2). As in the  $k=4$  historical analysis, threshold selection and source assessment are not independent. Bold marks cells with nonzero power and  $\text{FAR} \leq \alpha+0.02$ .

| corruption    | $\alpha$ | CRESS (historical) |              |       | Target-only anchor |              |       |
|---------------|----------|--------------------|--------------|-------|--------------------|--------------|-------|
|               |          | FAR                | power        | prec. | FAR                | power        | prec. |
| clean         | 0.05     | <b>0.053</b>       | <b>0.764</b> | 0.976 | 0.000              | 0.000        | –     |
|               | 0.10     | <b>0.110</b>       | <b>0.833</b> | 0.936 | 0.000              | 0.000        | –     |
|               | 0.20     | <b>0.216</b>       | <b>0.898</b> | 0.917 | <b>0.182</b>       | <b>0.903</b> | 0.928 |
| blur          | 0.05     | <b>0.050</b>       | <b>0.751</b> | 0.976 | 0.000              | 0.000        | –     |
|               | 0.10     | <b>0.110</b>       | <b>0.813</b> | 0.957 | 0.000              | 0.000        | –     |
|               | 0.20     | 0.221              | 0.900        | 0.917 | <b>0.199</b>       | <b>0.904</b> | 0.922 |
| bright/contr. | 0.05     | <b>0.051</b>       | <b>0.770</b> | 0.976 | 0.000              | 0.000        | –     |
|               | 0.10     | <b>0.107</b>       | <b>0.831</b> | 0.958 | 0.000              | 0.000        | –     |
|               | 0.20     | 0.223              | 0.895        | 0.916 | <b>0.185</b>       | <b>0.904</b> | 0.926 |
| Gauss. noise  | 0.05     | <b>0.035</b>       | <b>0.155</b> | 0.962 | 0.000              | 0.000        | –     |
|               | 0.10     | 0.138              | 0.398        | 0.951 | 0.000              | 0.000        | –     |
|               | 0.20     | 0.232              | 0.749        | 0.917 | 0.392              | 0.904        | 0.866 |
| JPEG          | 0.05     | 0.071              | 0.296        | 0.972 | 0.000              | 0.000        | –     |
|               | 0.10     | 0.141              | 0.588        | 0.959 | 0.000              | 0.000        | –     |
|               | 0.20     | 0.260              | 0.826        | 0.918 | 0.380              | 0.911        | 0.875 |

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**Table 11**

Historical comparison of smoothed conformal p-values and non-independent CRESS below the target-only floor ( $k=4$ , all classes, seeds 0–2, five pooled conditions). Randomized rates are exact expectations over the auxiliary randomization. Bold marks  $\text{FAR} \leq \alpha + 0.02$ .

| dataset | $\alpha$ | Randomized p-value |       | CRESS (historical) |       |
|---------|----------|--------------------|-------|--------------------|-------|
|         |          | FAR                | power | FAR                | power |
| MVTec   | 0.05     | 0.085              | 0.216 | <b>0.049</b>       | 0.216 |
|         | 0.10     | 0.170              | 0.431 | <b>0.105</b>       | 0.416 |
| VisA    | 0.05     | <b>0.036</b>       | 0.150 | <b>0.051</b>       | 0.144 |
|         | 0.10     | <b>0.072</b>       | 0.300 | <b>0.095</b>       | 0.351 |

**Table 12**

Historical matched-condition CRESS on 12 label-stratified VisA classes ( $k=4$ , seeds 0–2, five pooled conditions), using either within-VisA or MVTec source categories. Threshold selection and source assessment are not independent in this historical analysis. Bold marks cells with nonzero power and  $\text{FAR} \leq \alpha + 0.02$ .

| source pool           | $\alpha$ | FAR          | power        | prec. | no-harm |
|-----------------------|----------|--------------|--------------|-------|---------|
| VisA (within)         | 0.05     | <b>0.051</b> | <b>0.144</b> | 0.847 | 89%     |
|                       | 0.10     | <b>0.095</b> | <b>0.351</b> | 0.843 | 84%     |
|                       | 0.20     | <b>0.192</b> | <b>0.637</b> | 0.810 | 78%     |
| MVTec (cross-dataset) | 0.05     | <b>0.013</b> | <b>0.091</b> | 0.936 | 97%     |
|                       | 0.10     | <b>0.040</b> | <b>0.201</b> | 0.881 | 93%     |
|                       | 0.20     | <b>0.098</b> | <b>0.415</b> | 0.852 | 93%     |

**Table 13**

Protocol-locked representative comparison of fixed LOIO conformal and Vector Platt across cross-dataset, leave-one-category-out (LOCO), and within-split settings.  $\Delta\text{ECE}$  is LOIO minus Vector ECE, so negative values favor LOIO; no-harm is the fraction of cells in which LOIO does not increase ECE.

| Protocol     | Vector ECE | LOIO ECE     | $\Delta$ ECE | No-harm |
|--------------|------------|--------------|--------------|---------|
| MVTec→VisA   | 0.301      | <b>0.099</b> | −0.203       | 100%    |
| VisA→MVTec   | 0.304      | <b>0.102</b> | −0.202       | 100%    |
| LOCO         | 0.304      | <b>0.116</b> | −0.188       | 100%    |
| Within split | 0.361      | <b>0.154</b> | −0.207       | 100%    |

**Table 14**

MVTec pixel-level localization and FGSM fragility diagnostics, averaged over classes and five seeds. The upper block reports pixel AUROC and AU-PRO; the lower block reports clean and perturbed image AUROC under the label-aware PCA-residual surrogate.

| <i>Pixel localization</i>                      |                      |               |              |
|------------------------------------------------|----------------------|---------------|--------------|
| $k$                                            | Method               | pixel AUROC   | AU-PRO       |
| 1                                              | PCA64 (ours)         | <b>0.944</b>  | 0.820        |
|                                                | Controlled DINOv2 NN | 0.940         | <b>0.845</b> |
| 4                                              | PCA64 (ours)         | <b>0.957</b>  | 0.862        |
|                                                | Controlled DINOv2 NN | 0.951         | 0.862        |
| 8                                              | PCA64 (ours)         | <b>0.960</b>  | <b>0.876</b> |
|                                                | Controlled DINOv2 NN | 0.952         | 0.863        |
| <i>FGSM fragility diagnostic (image AUROC)</i> |                      |               |              |
| $\epsilon$                                     | $k$                  | clean → FGSM  | rel. drop    |
| 2/255                                          | 4                    | 0.937 → 0.158 | 83%          |
|                                                | 8                    | 0.945 → 0.175 | 82%          |
| 8/255                                          | 4                    | 0.937 → 0.446 | 52%          |
|                                                | 8                    | 0.945 → 0.446 | 53%          |

**Table 15**

Aggregation-sensitivity ablation: clean image AUROC averaged over representative classes and seeds 0–2 for max and top- $p$  mean pooling. Bold marks the best aggregator in each row.

| dataset | $k$ | max    | top-0.5%      | top-1% | top-2%        | top-5% |
|---------|-----|--------|---------------|--------|---------------|--------|
| MVTec   | 4   | 0.9571 | 0.9638        | 0.9667 | <b>0.9693</b> | 0.9621 |
|         | 8   | 0.9578 | 0.9685        | 0.9740 | <b>0.9759</b> | 0.9727 |
| VisA    | 4   | 0.9039 | <b>0.9101</b> | 0.9038 | 0.8944        | 0.8806 |
|         | 8   | 0.9101 | <b>0.9200</b> | 0.9147 | 0.9071        | 0.8941 |

**Table 16**

Novelty positioning: each ingredient has established prior art, and this paper’s contribution begins after it. The right column marks where prior work ends and our claims start.

| Work family                    | Established prior art                                                        | Our contribution starts after it                                                                                 |
|--------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| AnomalyDINO<br>SubspaceAD      | DINOv2 few-shot patch/memory-bank AD<br>DINOv2 PCA/subspace residual ranking | Reliability of the alarms such detectors raise<br>Operational reliability audit downstream of the fixed residual |
| Calibration/robustness studies | ECE, Platt scaling, FGSM fragility                                           | Conformal-native audit; full calibrator toolbox under one label-free protocol                                    |
| Conformal AD                   | P-values from nonconformity scores                                           | The few-shot floor, empirical source routing, and finite-category certificate feasibility                        |
| SAGE-style routing             | Gated shared/specialized experts                                             | Fixed no-label reliability routing and its failure boundaries                                                    |